FailureModeBench: A Unified Failure Taxonomy for Vision-Language Models

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Abstract

We introduce FailureModeBench, a benchmark and evaluation framework for diagnosing *how* vision-language models fail, not merely how often. The framework provides a modality-blind evaluation harness spanning 16 datasets across nine task families, supporting both image classification and visual question answering under a single prediction-record schema, so that errors arising from recognition and from reasoning become directly comparable. Every incorrect prediction is routed by an LLM judge into exactly one of eight failure modes (F1–F8), spanning visual and linguistic hallucination, cross-modal misalignment, fusion error, fabrication, misattribution, overgeneralization, and confabulation. An automated annotation pipeline produces a diagnostic corpus through deterministic stratified sampling, temperature-zero rubric judging, and content-hash de-duplication, yielding per-family failure-rate tables, per-class confusion matrices for fine-grained sets, and a dominant-mode summary. The benchmark evaluates VLMs from 4B open-weight to frontier proprietary systems over roughly 185K evaluated samples, under a fixed inference harness that holds prompting, label resolution, and quantization constant so that differences are attributable to the model rather than the scaffolding. **[TODO: headline result sentence, e.g. "even the best model achieves only X% on fine-grained recognition"] [TODO: failure-analysis sentence, e.g. "Y% of fine-grained failures stem from Overgeneralization rather than misperception"]** The framework and the benchmark are publicly available.

Introduction

Evaluating vision-language models has become an active area of research, with benchmarks targeting generic recognition (Deng et al., 2009), distribution-shift robustness (Hendrycks et al., 2021a; Recht et al., 2019), general multimodal understanding (Fu et al., 2023; Li et al., 2024), and visual mathematics (Lu et al., 2024; Zhang et al., 2024). However, most of these frameworks remain *score-centric*: they report one number per dataset and cannot explain the errors behind it. A model can score well on ImageNet while systematically collapsing subclasses into super-categories, or answer most of MathVerse while inventing intermediate steps on the rest.

Diagnosing VLM failure requires more than counting errors. The evaluator must determine where a response first departed from the evidence, distinguish a perceptual miss from a correct perception wrongly combined, separate an invented fact from an embellished one, and do so in a vocabulary that applies equally to a classifier naming a bird and a reasoner reading a chart. This makes failure diagnosis a distinct evaluation capability: the harness must not only record what the model got wrong, but characterize *how* the response broke.

Among the few efforts that look past aggregate accuracy, most target a single failure mode in isolation—object hallucination in captioning (Li et al., 2023; Rohrbach et al., 2018), or compositional relations (Wang et al., 2024b)—or operate on a single task format. Slice-discovery methods (Eyuboglu et al., 2022; Wu et al., 2019) identify *where* a model underperforms but not *how* it breaks. Moreover, existing diagnostic efforts typically cover one modality: a taxonomy built for captioning cannot be compared against one built for classification. As a result, there is no unified framework for systematically studying VLM failure structure across recognition and reasoning under a shared taxonomy.

To close this gap, we introduce **FailureModeBench**, a diagnostic framework for characterizing VLM failures. FailureModeBench extends classification and VQA evaluation with a shared record schema, an error-extraction stage, and a rubric-based failure judge, and delegates VQA execution to VLMEvalKit so that its scores remain comparable to published numbers. Figure 1 provides an overview of our system design.

Built on top of this framework, we construct a diagnostic benchmark designed to capture what prior work misses. The benchmark spans 16 datasets across ten task families and is organized along three orthogonal axes: two evaluation modalities, nine task families spanning perception to reasoning, and an eight-category failure taxonomy that partitions every error. The resulting benchmark covers roughly 185K evaluated samples, requiring models to be diagnosed under a fixed harness across a wide capability range. To make annotation scalable while maintaining quality, we develop an automated judging pipeline that produces the diagnostic corpus through deterministic stratified sampling, rubric-constrained single-label judging, and resumable incremental writes, reducing the annotation burden from full

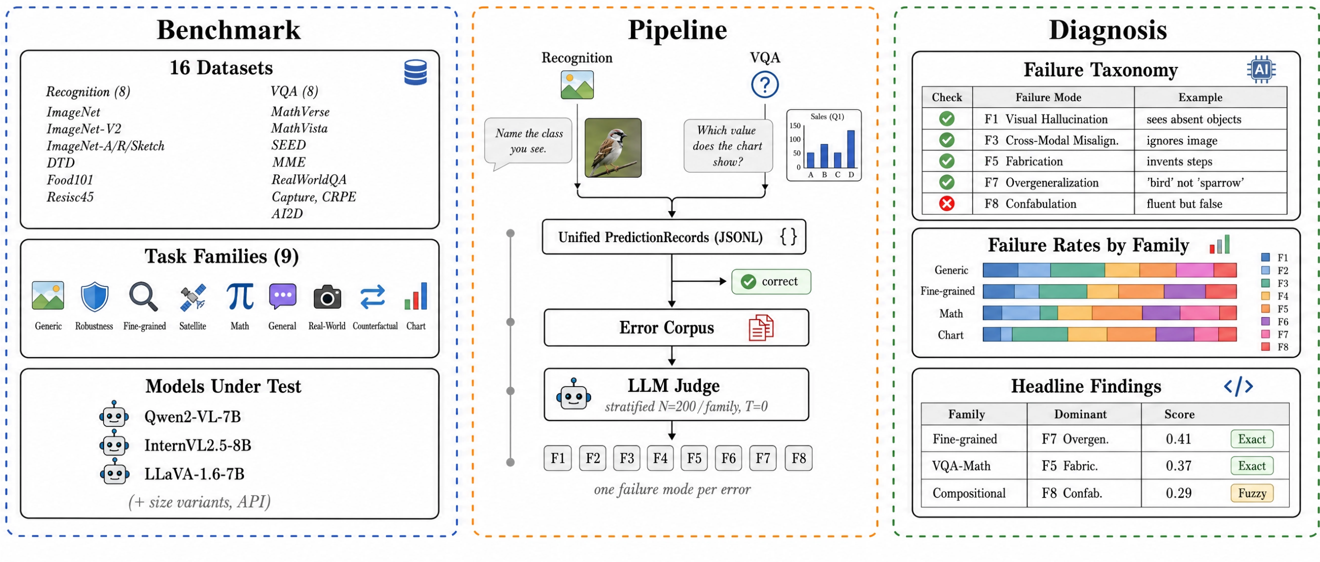

Figure 1: Overview of FailureModeBench. **Left (Benchmark):** 16 datasets spanning eight recognition and eight VQA sets, grouped into nine task families, evaluated over a default matrix of open-weight models plus size variants and a closed API model. **Center (Pipeline):** a recognition runner and a VQA runner emit an identical unified prediction record; correct predictions are set aside and only errors enter the corpus, which a stratified, temperature-zero LLM judge routes into exactly one of eight failure modes (F1–F8) per error. **Right (Diagnosis):** the failure taxonomy, per-family failure-rate distributions, and the headline table reporting each family's dominant mode. All downstream stages are modality-blind by construction, so a classification error and a chart-reasoning error are characterized under the same taxonomy. Values shown in the Diagnosis panel are illustrative of the reported format; measured results appear in Tables 3 and 4.

manual error analysis to targeted review. Finally, we conduct a systematic evaluation of open-weight and proprietary models and analyze their failure structure.

To summarize, our contributions are: 1. A unified framework and benchmark for diagnosing VLM failure across recognition and reasoning. 2. A scalable failure-annotation pipeline. 3. A systematic failure analysis of diverse models.

Related Work

Aggregate VLM benchmarks. A growing body of benchmarks evaluate VLMs on recognition and multimodal reasoning. ImageNet (Deng et al., 2009) and its variants—ImageNet-V2 (Recht et al., 2019), ImageNet-A (Hendrycks et al., 2021b), ImageNet-R and ImageNet-Sketch (Hendrycks et al., 2021a; Wang et al., 2019)—established the practice of measuring generalization by accuracy drop across distributions. MME (Fu et al., 2023), SEED-Bench (Li et al., 2024), and MMBench (Liu et al., 2024b) extend this to instruction-following VLMs, while MathVista (Lu et al., 2024) and MathVerse (Zhang et al., 2024) target visual mathematics. Despite this progress, these benchmarks report a single score per dataset and offer no account of the errors behind it.

Hallucination-specific evaluation. A parallel line targets one failure mode in isolation. POPE (Li et al., 2023) probes object hallucination through polar questions; CHAIR (Rohrbach et al., 2018) measures hallucinated objects in captions; CRPE (Wang et al., 2024b) probes compositional relations and the hallucinations they induce. These are sharp instruments, but each measures a single axis on a single task format, and their results do not compose into a model's overall failure profile.

Error analysis and slice discovery. Domino (Eyuboglu et al., 2022) and Errudite (Wu et al., 2019) surface coherent slices where a model underperforms. This is complementary to our aim: slice discovery localizes failure, whereas a taxonomy characterizes it. Our harness includes an experimental slice-discovery module, but the contribution here is the taxonomy and the protocol applying it uniformly.

LLM-as-judge. Using a strong model to grade outputs is now standard (Zheng et al., 2023), with known caveats around position and verbosity bias. Our use is narrower than open-ended preference scoring: single-label classification against a fixed rubric, with the image attached, at temperature zero.

A feature comparison with prior benchmarks is in Table 1.

Table 1: Comparison of representative VLM evaluation benchmarks. **Diagnostic** indicates whether errors are characterized beyond a correct/incorrect label. **Modality** distinguishes recognition (classification against a label set) from VQA (multiple-choice or open question answering); *both* denotes a shared record schema across the two. **Taxonomy** gives the number of failure categories reported, and **Unified** indicates whether one taxonomy spans all evaluated tasks.

Benchmark	Diagnostic	Modality	Taxonomy	Unified	Datasets
ImageNet variants (Hendrycks et al., 2021a)	×	recognition	—	×	5
MME (Fu et al., 2023)	×	vqa	—	×	1
SEED-Bench (Li et al., 2024)	×	vqa	—	×	1
MMBench (Liu et al., 2024b)	×	vqa	—	×	1
MathVista (Lu et al., 2024)	×	vqa	—	×	1
POPE (Li et al., 2023)	✓	vqa	1 (object hall.)	×	3
CHAIR (Rohrbach et al., 2018)	✓	captioning	1 (object hall.)	×	1
CRPE (Wang et al., 2024b)	✓	vqa	1 (relation hall.)	×	1
Errudite (Wu et al., 2019)	✓	text	user-defined	×	—
Domino (Eyuboglu et al., 2022)	✓	recognition	slices	×	—
FailureModeBench	✓	**both**	**8 (F1–F8)**	✓	**16**

FailureModeBench

FailureModeBench aims to measure *failure structure*: whether the errors of a model can be characterized, compared across task families, and attributed to a specific breakdown in the perception-to-answer pipeline. To this end, the framework adopts a lightweight, resumable, and reproducible design philosophy, extending conventional accuracy-based evaluation to diagnostic settings through explicit error handling. Together, these design choices make FailureModeBench a generic runtime for characterizing multimodal failure. Figure 1 provides an overview of our system design.

Architecture

Each run is parameterized by a *matrix*, which specifies the models under test, the datasets, the sampling budget for the judge, and the provider backing it. Evaluation then unfolds among a *model* (under test), a *judge*, and an *aggregator*. The model produces predictions; the judge characterizes the errors; the aggregator maintains the tables and figures that constitute the reported result.

The harness operates over two channels. The inference channel runs the model against a dataset and emits records; the diagnostic channel consumes only the subset of those records marked incorrect. The separation matters: the judge never sees correct predictions, so its cost scales with the error rate rather than with dataset size.

Record management. Both runners emit an identical typed record:

```
{sample_id, dataset, family,
modality, model, question, gold,
prediction, pred_label, correct,
image_ref, extra}
```

Because the schema is shared, every downstream stage is modality-blind: error extraction, sampling, judging, aggregation, and figure generation are written once and apply to a classification error and a chart-reasoning error alike. Only errors retain a saved image, which bounds disk usage on the large ImageNet splits.

Failure Diagnosis

Failure diagnosis is the main focus of this work. We treat errors as first-class artifacts rather than decrements in a score. Each error is assigned a stable content hash, stored in a corpus, and referenced by the judge and the aggregator in the same way that predictions are referenced. This design makes error annotation persistent, resumable, and traceable across runs: a rate-limited or crashed judging pass resumes exactly where it stopped, and re-running an unchanged matrix reproduces the corpus deterministically.

Evaluation Modes

FailureModeBench implements two inference modes. In *Recognition* mode, the model names the class in free text and the answer is resolved against a committed label set by an ordered cascade: exact match, synonym lookup, token-level Jaccard overlap, and optionally semantic embedding similarity. The cascade is ordered so that high-precision rules fire first and the fuzzy matcher is reached only on their failure, which prevents label resolution from silently manufacturing correctness. This interface naturally supports per-class confusion analysis, with the drawback that a resolution miss is scored as a model error.

In *VQA* mode, the model is executed through VLMEvalKit, which owns loaders, prompt templates, and answer extraction. We import its per-sample prediction spreadsheet and carry over its correctness column rather than re-deriving it. This interface keeps our scores comparable to published numbers and is robust to extraction edge cases, but cedes control over prompting to the external harness.

Preprocessing. One rule applies uniformly across both modes: images with any side exceeding 1000 px are dropped. This bounds vision-token cost for dynamic-resolution models and keeps memory predictable on a single consumer GPU. The filter and the resulting counts are asserted at aggregation time, so Table 2 cannot silently drift from the code.

Evaluation

FailureModeBench evaluates models through two complementary mechanisms. *Score-based evaluation* compares predictions against gold labels, providing an objective, reproducible outcome signal: top-1 accuracy for recognition, VLMEvalKit's score for VQA. *Rubric-based failure judging* assesses each error's character, assigning one of F1–F8 with a rationale and a confidence. Together, these mechanisms cover both the outcome and the structure of each run. Concrete metrics are defined in the Benchmark section.

Benchmark

The FailureModeBench benchmark comprises 16 datasets spanning diverse visual domains, reasoning patterns, and error opportunities. Each dataset is assigned to a task family that determines how its errors are pooled for diagnosis. We describe the design dimensions that structure the benchmark, followed by its composition and metrics.

Design Dimensions

We structure the benchmark along three orthogonal axes—modality, task family, and failure mode—that jointly control what is measured, how errors are pooled, and how they are characterized.

Modalities. Two modalities determine the answer format and the scoring rule. *Recognition* constrains the model to a label set and admits confusion analysis; *VQA* admits free-form or multiple-choice answers and admits reasoning-chain analysis. The distinction is not cosmetic: a constrained classifier cannot fabricate a reasoning step, so the two modalities are expected to occupy different regions of the taxonomy, and measuring that divergence is the point of sharing one.

Task families. The family prescribes the perceptual and reasoning demand connecting the image to the answer. We derive nine families by decomposing the space along the perception-to-reasoning axis: *generic* recognition over natural images; *robustness* under adversarial, rendition, and sketch shift; *fine-grained* recognition requiring subclass discrimination; *satellite* imagery with non-canonical viewpoints; *mathematical* reasoning over diagrams; *general understanding* of scenes, including compositional relations and the hallucinations they induce; *real-world* question answering; *counterfactual* understanding; and *chart understanding*. These families impose genuinely different demands: fine-grained recognition stresses discrimination within a super-category, while chart understanding stresses the composition of correctly perceived parts.

Failure modes. The taxonomy prescribes how each error is characterized. Every incorrect prediction receives exactly one mode. The single-label constraint is deliberate: it forces a partition rather than a set of overlapping tags, which makes per-family distributions interpretable as percentages summing to one hundred. Where an error admits two readings, the rubric instructs the judge to select the mode describing the *earliest* departure from the evidence—a perceptual error propagating into a wrong computation is F1 or F3, not F4. The eight modes are:

F1 — Visual Hallucination. Reports visual entities or features absent from the image; the failure originates in perception. Characteristic of ImageNet-A, where the model "sees" structure in misleading texture.

F2 — Linguistic Hallucination. Unsupported verbal claims grounded neither in image nor question. The perceptual content may be correct; the language layer adds unlicensed attributes.

F3 — Cross-Modal Misalignment. Ignores the image or mis-binds text to it. Distinguished from F1 in that nothing is invented—the answer is a plausible label drawn from the language prior rather than from the image.

F4 — Fusion Error. Every part is read correctly and combined wrongly: the compositional failure. Diagnostic on AI2D and MathVista.

F5 — Fabrication. Whole facts, steps, or relationships are invented. Where F2 embellishes, F5 constructs—a MathVerse formula step existing nowhere in the problem.

F6 — Misattribution. The property identified is correct but assigned to the wrong entity or region: right content, wrong binding.

F7 — Overgeneralization. A super-category where a subclass was required. The mode most invisible to aggregate accuracy, since it scores identically to a wild confusion despite reflecting partially correct perception.

F8 — Confabulation. A fluent, internally coherent, false reasoning chain—a plausible narrative on MME, or an inverted relation on CRPE stated with confidence.

Table 2: The 16 datasets of FailureModeBench. "Used" is the count after the 1000 px resolution filter; "Orig." is the released split size.

Family / Dataset	Focus	Used	Orig.
Recognition			
ImageNet	Generic	49,032	50,000
ImageNet-V2	Generic	9,772	10,000
ImageNet-A	Generic (adv.)	7,467	7,500
ImageNet-R	Texture	28,506	30,000
ImageNet-Sketch	Edges	35,350	50,000
DTD	Edges, Texture	5,640	5,640
Food101	Fine-grained	25,250	25,250
Resisc45	Satellite	4,500	4,500
VQA			
MathVerse (MCQ)	Mathematical	1,631	2,180
MathVista	Mathematical	490	1,000
SEED	General underst.	3,881	13,991
MME	General underst.	1,576	2,370
RealWorldQA	Real-world	765	765
Capture	Counterfactual	817	962
CRPE	General (comp.)	7,575	7,575
AI2D	Chart & graph	2,704	3,090
Total		**184,956**	**214,823**

The definitions and the judge rubric live in a single module shared by the prompt builder and the response parser, so adding or renaming a mode updates both together. This removes the most common source of taxonomy drift: a prompt and a parser that disagree about the categories. The Diagnosis panel of Figure 1 shows the taxonomy alongside the per-family failure distributions and dominant-mode table it feeds.

Benchmark Composition

Dataset sources. We source evaluation data from 16 established datasets spanning complementary visual domains: generic natural images (ImageNet (Deng et al., 2009), ImageNet-V2 (Recht et al., 2019)), distribution shift (ImageNet-A (Hendrycks et al., 2021b), ImageNet-R (Hendrycks et al., 2021a), ImageNet-Sketch (Wang et al., 2019)), textures (DTD (Cimpoi et al., 2014)), fine-grained categories (Food101 (Bossard et al., 2014)), remote sensing (Resisc45 (Cheng et al., 2017)), visual mathematics (MathVerse (Zhang et al., 2024), MathVista (Lu et al., 2024)), general understanding (SEED (Li et al., 2024), MME (Fu et al., 2023)), real-world QA (RealWorldQA), counterfactual understanding (Capture), compositional relations (CRPE (Wang et al., 2024b)), and diagrams (AI2D (Kembhavi et al., 2016)). This diversity ensures that a reported failure profile reflects general model behavior rather than proficiency on a single image type. Table 2 gives the composition.

Models under test. The default matrix comprises Qwen2-VL-7B-Instruct (Wang et al., 2024a), InternVL2.5-8B (Chen et al., 2024), and LLaVA-1.6-7B (Liu et al., 2024a). Two size-variant pairs support scale ablation: InternVL2.5-4B against 8B, and LLaVA-1.6-7B against 13B, isolating scale within an architecture family. A closed API model is supported through the same backend interface, so a frontier system can be compared against open weights without a second code path.

Evaluation Protocol

Inference implementation. To isolate model capability from scaffolding, all models share a single fixed harness. Every model receives the same recognition prompt, the same label cascade, and the same VLMEvalKit configuration. The reference hardware is a single 11 GB RTX 2080 Ti; the 7B, 8B, and 13B models load in 4-bit, and Qwen2-VL's maximum vision-token count is capped to bound its dynamic-resolution memory growth. We report quantization as part of the configuration rather than as an implementation detail: it can plausibly shift the failure profile even where it barely moves accuracy. We hold this harness fixed on purpose—comparing models under a common, well-specified harness is what makes differences attributable to the model rather than to the surrounding system.

Metrics. *Accuracy* is the outcome metric: top-1 for recognition, VLMEvalKit's score for VQA. *Failure distribution* is the primary diagnostic metric: for each family, the percentage of judged errors falling in each of F1–F8, with the dominant mode reported per family. *Macro-F1 and confusion matrices* give per-class resolution on the fine-grained sets, distinguishing a model that spreads errors uniformly from one that collapses a specific subclass pair.

Judge. By default we use Claude as the failure judge, at temperature zero. Per family, a deterministic stratified sample of up to N errors (default 200) is drawn, balanced across (model, dataset) cells so that a single large dataset cannot dominate a family's profile. The judge receives the rubric, the question, the gold answer, the prediction, and—by default—the image, and returns one mode with a rationale and a confidence. Unparseable outputs are retried three times and then recorded as `NA` rather than forced into a category. The corpus is de-duplicated by content hash and written incrementally, making resumption idempotent by construction.

Annotation Pipeline

The pipeline operates in four stages, producing the diagnostic corpus described above.

Stage 1: Recognition inference. Each split is streamed, the resolution filter applied, and the model prompted to name the class. The free-text answer is resolved through the ordered cascade; only errors retain a saved image.

Stage 2: VQA inference. VLMEvalKit is invoked per (model, dataset) pair and its prediction spreadsheet imported into the unified schema. The importer tolerates column-name drift across VLMEvalKit versions and carries over the external correctness determination, so our VQA numbers remain comparable to published ones.

Stage 3: Failure judging. All errors are gathered into a corpus and stratified per family. Each sampled error is judged against the shared rubric at temperature zero, producing one mode, a rationale, and a confidence.

Stage 4: Aggregation. The final stage emits accuracy per (model, dataset); failure distributions by family, dataset, and model; per-class confusion matrices for the fine-grained sets; the figures; and a summary recording the dominant mode per family. The by-family table is the headline artifact.

Quality Verification

Offline self-test. An end-to-end offline test validates the pipeline without a GPU or an API key: it confirms that error gathering and stratified sampling execute, that a stub judge's verdicts all parse, that judge resumption is idempotent, that the aggregation tables and confusion matrices are written, and that both figures render headlessly. The recognition data path has additionally been exercised on real DTD data.

Human agreement study. **[TODO: Run the agreement study: sample N judged errors, have an expert annotator re-label them blind to the judge's verdict, and report raw agreement and Cohen's κ over the eight modes. This is the single most important missing validation—the F1–F8 distributions are only as trustworthy as the judge's agreement with a human, and an 8-way κ is a substantially harder target than the binary pass/fail agreement reported by judge-based agent benchmarks.]**

Experiments

[TODO: This section reports the full matrix run. Execute `RUN=main bash scripts/run_rolf.sh`, then populate from `tables/accuracy.csv` and `tables/failure_by_family.csv`.]

We evaluate **[TODO: M]** vision-language models on FailureModeBench, spanning proprietary and open-weight systems across a wide parameter range. Unless otherwise noted, all models are evaluated under the fixed harness described above, with 4-bit quantization and the 1000 px resolution filter. The metrics are accuracy, per-family failure distribution, and macro-F1 on the fine-grained sets.

Table 3: Main results. All models evaluated under the fixed harness with 4-bit quantization. **[TODO: populate from tables/accuracy.csv]**

Model	Recog. Acc. ↑	VQA Score ↑	Macro-F1 ↑
InternVL2.5-4B	--.-	--.-	--.-
LLaVA-1.6-7B	--.-	--.-	--.-
Qwen2-VL-7B	--.-	--.-	--.-
InternVL2.5-8B	--.-	--.-	--.-
LLaVA-1.6-13B	--.-	--.-	--.-
Claude (API)	--.-	--.-	--.-

Main Results

The main experimental results are presented in Table 3. We highlight four observations:

1. [TODO: Aggregate accuracy hides failure structure.] [TODO: Contrast two models with comparable accuracy but divergent failure profiles—this is the core motivating claim. If no such pair exists in the results, say so explicitly; it weakens the framing and should be reported.]

2. [TODO: Recognition and reasoning occupy different regions of the taxonomy.] [TODO: Report whether recognition errors concentrate in F3/F6/F7 and VQA errors in F4/F5/F8, as the taxonomy in Figure 1 anticipates. Quantify the overlap.]

3. [TODO: Scale shifts the failure profile.] [TODO: Use the InternVL2.5 4B/8B and LLaVA-1.6 7B/13B pairs. Report whether the dominant mode changes with scale within an architecture family, or only its magnitude.]

4. [TODO: Overgeneralization is invisible to accuracy.] [TODO: Report the F7 share on the fine-grained and satellite families: the fraction of errors reflecting partially correct perception, scored identically to wild confusions.]

Failure Analysis

Beyond aggregate metrics, we perform a detailed failure analysis to understand *why* models fail and where capability improvements are most needed. Our rubric judge assigns each error exactly one of F1–F8; by examining which modes dominate each family, we construct a failure profile per family and per model. Table 4 reports the distribution.

[TODO: Write the analysis once the table is populated. The claims the taxonomy is built to adjudicate—stated here as hypotheses, to be confirmed or refuted: (i) fine-grained recognition errors are dominated by Overgeneralization (F7) and Cross-Modal Misalignment (F3),

Table 4: Failure-mode distribution by task family. Percentages are relative to each family's judged errors and sum to ≈100 per row. **[TODO: populate from tables/failure_by_family.csv]**

Family	n	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**	**F8**
Generic	–	–	–	–	–	–	–	–	–
Robustness	–	–	–	–	–	–	–	–	–
Fine-grained	–	–	–	–	–	–	–	–	–
Satellite	–	–	–	–	–	–	–	–	–
Math	–	–	–	–	–	–	–	–	–
General	–	–	–	–	–	–	–	–	–
Real-World	–	–	–	–	–	–	–	–	–
Counterfact.	–	–	–	–	–	–	–	–	–
Chart	–	–	–	–	–	–	–	–	–

i.e. the model perceives correctly but answers at the wrong granularity, or ignores the image in favor of the prior; (ii) VQA errors cluster in Fabrication (F5) and Confabulation (F8), i.e. the model produces fluent reasoning unanchored to the evidence; (iii) the fine-grained confusion matrices show structured rather than uniform confusion. If the data refute these, report the refutation—a taxonomy that only confirms its designers' expectations is not measuring anything.]

Ablations

[TODO: Label-matching ablation: run with and without `--embeddings` and report how much of the recognition error rate is genuine model error versus label-resolution failure. This bounds a known confound and should be reported before the main accuracy numbers are trusted.]

[TODO: Text-only judging ablation: run the judge with `--no-images` and compare mode assignments against the image-attached default. Divergence quantifies how much of the judge's verdict depends on its own perception rather than on the text of the prediction—directly relevant to whether F1 verdicts are trustworthy.]

[TODO: Sampling-budget ablation: vary `--n-per-family` and report the budget at which the dominant mode per family stabilizes.]

Conclusion

We introduced FailureModeBench, a unified framework and benchmark for diagnosing vision-language model failures. The framework provides a modality-blind evaluation harness spanning 16 datasets across nine task families, with a shared prediction-record schema that makes recognition and reasoning errors directly comparable. The accompanying taxonomy contributes eight failure modes partitioning every error, applied by a resumable annotation pipeline that reduces the burden from full manual error analysis to targeted review. **[TODO: headline finding]**

Our analysis surfaces several observations that we hope will guide future work. **[TODO: Populate after the run. Candidate structure, mirroring the strongest form of this argument: first, the primary bottleneck claim—which failure mode dominates for capable models, and whether it is perceptual or compositional; second, whether failure modes shift with scale, implying different research directions at different capability levels; third, what the fine-grained confusion structure reveals that accuracy does not.]**

Limitations

We discuss the following limitations of our work. Our diagnostic metric relies on an LLM judge, and inherits its biases. The single-label constraint that makes the distributions interpretable also forces a choice on genuinely ambiguous errors; the earliest-departure tie-break makes that choice consistent but not necessarily correct. The judge's own perceptual limits bound the annotation: an F1 verdict requires determining that a feature is *absent* from an image, which is itself a VLM task at which the judge may fail in the same ways as the model under test. **[TODO: report human agreement and κ once measured; if the judge is a Claude model and a Claude model is also under test, note the self-preference confound explicitly (Zheng et al., 2023)]**

Judging up to 200 errors per family bounds cost but means family-level distributions carry sampling error, and the (model, dataset) balancing that prevents domination by large datasets simultaneously prevents the profile from reflecting the natural error mixture—the right trade for diagnosis, the wrong one for estimating a population failure rate. **[TODO: report whether results are single-run, and quantify run-to-run variance]**

The 1000 px filter removes a small but non-random slice—plausibly the highest-resolution, most detail-dependent items, precisely where fine-grained failures might concentrate. Recognition is scored through free-text label resolution, so a resolution miss is scored as a model error even where the model was right in substance; the ordered cascade limits but does not eliminate this. Two VLMEvalKit identifiers require verification against the installed version, and the counterfactual set remains a placeholder pending confirmation.

Finally, eight modes are a compression of a richer failure space, and the taxonomy is defined over errors only: it says nothing about correct answers reached by faulty reasoning, which a confabulation-focused reading would want to capture. To keep model comparisons fair, we evaluate every model under a single fixed harness rather than tuning prompts per model; our numbers therefore reflect model behavior under this standardized setup rather than each model's best achievable performance.

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